

Quiz 4: Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. You may use the results covered in Lectures 1–5. When a proof directly from the definition is requested, use the epsilon–delta definition. All functions are real-valued. Limits and continuity are considered within the stated domain; at an included endpoint of an interval, continuity is one-sided.

1. Find all $m, b \in \mathbb{R}$ for which the function

$$f(x) = \begin{cases} x^2, & x \leq 0, \\ mx + b, & 0 < x < 2, \\ \sqrt{x+7}, & x \geq 2 \end{cases}$$

is continuous on $\mathbb{R}$.

Solution.

Continuity at zero requires $b = 0$, since the left-hand limit and value are zero and the right-hand limit is b . At 2, the left-hand limit is $2m + b$, whereas the right-hand limit and value are 3. Hence $2m + b = 3$, giving

$$\boxed{m = \frac{3}{2}, \quad b = 0.}$$

Conversely, these values match both joins. Each branch is continuous on its own interval, so the resulting function is continuous on $\mathbb{R}$.

2. Let I be a nonempty interval and let $f : I \rightarrow \mathbb{R}$ be continuous. Suppose $f(x)^2 = 1$ for every $x \in I$. Prove that either $f(x) = 1$ for every $x \in I$, or $f(x) = -1$ for every $x \in I$.

Solution.

The equation $f(x)^2 = 1$ forces each value to be 1 or -1 . If both values occurred, there would be two points $u < v$ in I with opposite function values. Since I is an interval, it contains $[u, v]$. Continuity and the Intermediate Value Theorem would give a $c \in (u, v)$ with $f(c) = 0$, contradicting $f(c)^2 = 1$. Thus both signs cannot occur. Since I is nonempty, all values are 1, or all are -1 .